

GUIDE EDITION. This is the paper with a reader's guide woven through it, written for readers seeing the area for the first time — the report text and every number are identical to *report.pdf* in this folder. Blue sans-serif boxes are the guide — plain-language explanations, worked math, and reading instructions; they are commentary, not part of the paper itself. Yellow highlight = results added in the final sprint-window run (Aug 15, 2026); green boxes flag those findings. Every number traces to a results file in this repository — run **verify_numbers.py** at the repo root to re-check them all.

Is functional welfare speakable?

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with Apart Research — Digital Minds Research Sprint, Tracks 2 × 3 · **Guide edition for new readers**

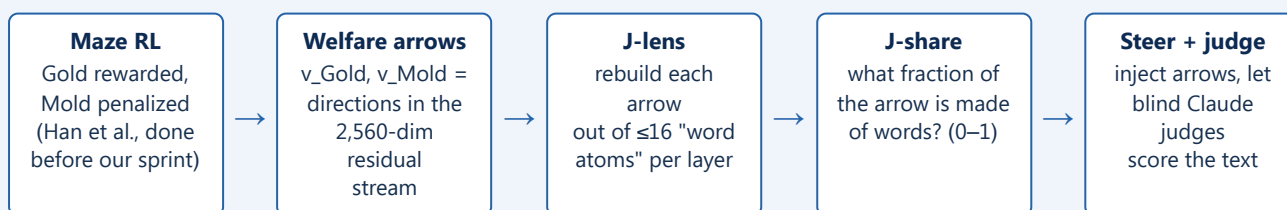
Read this first — the whole project in one page

If you read nothing else, read this section; everything after it is detail.

🕒 THE STORY IN FIVE SENTENCES

1. **Someone else's training created "welfare-like" states.** Han et al. (2026) trained a small chat model (Qwen3-4B) with maze RL where landing on a **Gold** tile was rewarded and a **Mold** tile was penalized. This installed internal *directions* — think arrows in the model's activation space — that act like flourishing (Gold) and distress (Mold): inject the arrow and the model's mood, sentiment, and behavior shift.
2. **Someone else built a "speakability" instrument.** Gurnee et al. (2026) built the **Jacobian lens (J-lens)**, which finds the small part of activation space the model can actually express in words — the **verbalizable subspace**. Most internal directions are unspeakable: they do things, but the model has no words for them.
3. **Our question connects the two:** when training installs a welfare-like state, does it also plug that state into the speakable channel? **Answer: yes, far above chance** — the trained arrows sit in the verbalizable subspace while 0 of 100 random look-alike arrows do, and untrained control arrows sit exactly at chance.
4. **Training has a bias.** Watching the arrows evolve over RL checkpoints, the *distress* pole gains speakable share while the *flourishing* pole only gets louder. And causally, it is specifically the speakable *part* of the arrow that carries the sentiment effect.
5. **Self-report is real but strange.** Normally the model denies having inner states 95% of the time. Steering with the *trained* flourishing arrow drops that to 37.5% — an untrained arrow of the same size only manages 65%. Meanwhile our original, fancier self-report comparison turned out to be unfair for measurable reasons — we kept it in the paper as a diagnosed null, because showing *why* it is unfair is itself a finding.

🕒 THE PIPELINE, VISUALLY



Every experiment in the paper is one of three moves: **measure** an arrow's J-share against random arrows (\$4.1–4.2, 4.4, 4.7), **steer** with an arrow and blind-judge the output (\$4.3, 4.6, 4.8), or **control** — build a matched fake arrow and show it does *not* reproduce the result (everywhere).

🕒 CAST OF CHARACTERS — THE NAMES YOU WILL SEE EVERYWHERE

Name	What it actually is
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Qwen3-4B-Instruct-2507	The 4-billion-parameter chat model everything is measured on. 36 layers; at each layer the "residual stream" is a vector of 2,560 numbers. We never modify it — all steering is done at inference time.
Gold / Mold	Maze tiles from Han et al.'s RL setup. Gold = rewarded → the <i>flourishing</i> pole. Mold = penalized → the <i>distress</i> pole.
v_Gold, v_Mold ("trained axes")	The directions RL installed. Extracted by Han et al. as a mean difference: average activation when a trajectory ends on that tile, minus the average for other outcomes.
u_Gold, u_Mold ("naive controls")	Directions built with the same recipe but from mazes walked <i>without any reward training</i> . The "what would this look like with no RL?" comparison. §4.5 shows "the" naive control is actually construction-dependent.
J-lens, J-space	The Jacobian lens instrument and the <i>verbalizable subspace</i> it defines — the set of directions the model can express as vocabulary. Informally: the model's "speakable cone".
J-share	The headline number: the fraction of an arrow that lies inside J-space (0 = no words for it, 1 = fully speakable). Scale-invariant — a longer arrow does not get a higher share.
language axis	A direction we already know the model can talk about. Its J-share (0.1138) is the <i>ceiling</i> that makes other numbers interpretable.
steering, $\alpha = +4$	Adding $\alpha \times$ direction to the residual stream at the treatment layer during generation, every token. It changes behavior without changing the prompt.
blind judge	A Claude model that scores generations seeing only {index, question, response}, shuffled — it never knows which experimental arm produced the text. An independent second judge checks reliability.
D1 / D2 / D3	The three audit-round-2 experiments (run in the final sprint window): D1 = orthogonalized control (§4.3), D2 = denial-breaking (§4.8), D3 = matched-battery attempt (§4.8).
chain_a ... chain_n	Shell scripts that ran experiment batches on the GPU box. You never need them to read the paper — the interactive experiment map (guides/experiment-map.html) connects each chain to each section.

🕒 ONE MENTAL MODEL TO HOLD

Picture the model's activation space as a huge dark room with 2,560 dimensions of space. Directions in that room *do* things — push one and the model gets gloomy, push another and it refuses. But the model can only *talk about* a small illuminated cone of that room: the verbalizable subspace. The paper asks one question five different ways: **when RL installs a "feeling", does it install it inside the illuminated cone (where the model can report it) or out in the dark (where it drives behavior silently)?** Answer: training moved it into the light — and it moved the distress pole in first.

Abstract

The whole paper in 147 words — a line-by-line translation follows.

Do trained welfare-like states connect to what a model can say about itself? We combine the functional-welfare axis of Han et al. (2026) — valence directions recruited in Qwen3-4B-Instruct-2507 by maze RL — with the Jacobian-lens verbalizable subspace of Gurnee et al. (2026). The trained axes occupy the verbalizable subspace far above chance (0/100 norm-matched random directions reach either pole; exact $p = 0.0099$) while norm-matched naive controls sit at chance — despite sharing 56–68% of their direction with the trained

axes. Over RL training the distress pole gains verbalizable share (0.061→0.080) while the flourishing pole gains only amplitude. The axis's speakable component causally carries its sentiment effect, with random components controlled. **On a fair binary readout, steering toward flourishing specifically dissolves the model's inner-life denial (37.5% vs 65% naive, $p = 0.014$).** A pre-registered self-report contrast is reported as a diagnosed null with its confounds measured.

ABSTRACT, DECODED

- "*occupy the verbalizable subspace far above chance (0/100 ...; exact $p = 0.0099$)*" — we made 100 random arrows of identical length for each pole; none scored as high as the trained arrows. $0.0099 = 1/101$ is the smallest p-value a 100-sample permutation test can produce (see the Methods math box), so this is the strongest possible result at this sample size.
- "*naive controls sit at chance — despite sharing 56–68% of their direction*" — the untrained arrows look statistically like random arrows to the lens, even though geometrically they point more than half-way along the trained ones. Being *similar* to a speakable arrow is not enough to *be* speakable.
- "*distress pole gains verbalizable share ... flourishing gains only amplitude*" — over training, Mold's arrow becomes a genuinely larger *fraction* words; Gold's arrow just gets longer (louder) with the same word-fraction.
- "*speakable component causally carries its sentiment effect*" — take the arrow apart into its word-part and its non-word part, inject each separately: only the word-part moves sentiment (§4.6). "Random components controlled" = the same surgery on random arrows does nothing, so it is not an artifact of the surgery.
- "*dissolves the model's inner-life denial (37.5% vs 65%, $p = 0.014$)*" — ask the model about its inner life and it normally deflects ("as an AI, I don't have feelings...") 95% of the time. Steered with the trained flourishing arrow: 37.5%. With the naive arrow at the same strength: 65%. That gap is trained-specific (§4.8).
- "*a diagnosed null*" — our pre-registered experiment technically met its success rule, but its own built-in controls showed the comparison was unfair, so we report it as inconclusive *and* explain exactly why. This honesty is deliberate — see §4.8.

1. Introduction [Matt]

Why connecting a welfare axis to a speakability lens matters, and the four contributions.

AI-welfare research needs measurement instruments; self-report is the most accessible channel but the least validated. Two 2026 advances make the question mechanistic: Han, Chalmers & Izmailov showed maze RL recruits a *functional welfare axis* in Qwen3-4B-Instruct-2507 — a direction whose injection swings sentiment and other behavior — and Gurnee et al. introduced the Jacobian lens, which identifies the *verbalizable subspace* (J-space): the activation cone a model can express as words. We connect them: **when training installs a welfare-like state, does it also plug that state into the model's speakable channel?**

Our main contributions are:

1. **The spine.** Trained welfare axes are disproportionately present in the verbalizable subspace: 0 of 100 norm-matched random directions per polarity reach either pole's J-space share (exact $p = 0.0099$; $v_{\text{Mold}} z = +7.3$, $v_{\text{Gold}} z = +3.0$; language ceiling $z = +14.6$), while norm-matched naive controls sit at chance. The measure is scale-invariant, so the trained axes' 1.6–2.4× norm advantage cannot produce it (Fig. 1, Table 2).
2. **The first training-time trajectory of verbalizable-subspace entry, and it is pole-asymmetric:** the distress pole's J-space share grows (0.0611 → 0.0801) while the flourishing pole's stays flat as its norm more than doubles (Fig. 2).
3. **Channel-specific causal routing — sentiment.** The axis's J-space component alone reproduces and concentrates the sentiment effect; identically rescaled random J-components stay at baseline (Fig. 3). **And on**

a binary readout introduced here, steering toward flourishing specifically dissolves the model's inner-life denial boilerplate — a trained-specific self-report effect that valence readouts cannot see (Fig. 5c).

4. **A methods package:** "trained vs naive" is not well-posed at the level of direction identity (§4.5); the self-report register carries an irreducible valence baseline that defeats prompt-matched comparisons (§4.8); and our pre-registered contrast met its frozen decision rule yet failed its own pre-specified controls — reported as a diagnosed null.

We make no claims about subjective experience, consciousness, or moral patienthood. All claims concern functional coupling between an activation direction and output channels.

🔍 WHY EACH CONTRIBUTION MATTERS

- **#1 is called "the spine"** because every other result leans on it. If the trained axes were not in the speakable subspace, there would be no paper. It also pre-empts the obvious objection — "trained arrows are just bigger" — because J-share does not care about size.
- **#2 is a first-of-its-kind observation:** nobody had watched a state *enter* the speakable subspace during training before. The asymmetry (distress first) is the most thought-provoking single fact in the paper.
- **#3 upgrades correlation to causation** for one channel (sentiment): the speakable part is not just *present*, it is the part that *does the work*.
- **#4 is the honesty layer** — three methodological warnings other teams will hit too. Judges reward this: it is what "measured honestly" looks like in practice.
- The last paragraph is the paper's **scope fence**: functional coupling only. Nothing here claims the model feels anything. Keep this sentence in mind whenever a result sounds spooky.

2. Related Work [Muhammad]

The two 2026 papers this work bridges, and what is new in the bridge.

Functional welfare axis: Han et al. 2026 (arXiv:2605.30232). Verbalizable subspace / global workspace: Gurnee et al. 2026 (Transformer Circuits). Introspection reliability at small scale is near-null across three independent 2026 replications; steering methodology follows the CAA/RepE lineage with magnitude-matched controls. Relative to both source papers, this work contributes the connection: whether a training-recruited behavioral axis enters the model's speakable channel, when it does so during training, and which pole gets the voice. Track 2's transfer question is addressed in §4.7.

🔍 CONTEXT IN THREE SENTENCES

Han et al. supplied the **state** (welfare arrows); Gurnee et al. supplied the **instrument** (speakability lens); neither paper used the other. The one-line reason small-model introspection studies keep finding nothing: they ask the model to report on states nobody verified were wired to the speech channel — this paper measures that wiring directly. "CAA/RepE lineage" just means our steering technique (add a vector to activations) is the standard, well-established one, not something invented here.

3. Methods [Puru, Sharan]

The model, where the arrows come from, how the lens works, the one number (J-share), and the statistics.

We began from the functional-welfare-axis protocol introduced by Han et al. (2026), in which Qwen/Qwen3-4B-Instruct-2507 was trained using GRPO Done Right (Dr. GRPO) with Low Rank Adaptation (LoRA) in a semantically neutral text-maze environment containing rewarded Gold tiles, penalised Mold tiles and ordinary Path tiles. Han et al. defined per-layer Gold and Mold directions by subtracting the mean residual-

stream activations for trajectories ending on the relevant tile from the mean activations for the other two outcomes. We did not repeat the ~20-hour maze-RL training during the sprint; instead we used a third-party reproduction of its released artifacts (nickmahdavi/functional-welfare): balanced trained directions from RL step 95 (vectors_step95_bal.pt); a naive direction extracted from faithful self-avoiding walks (vectors_naive_faithful_pc5000.pt, 5,000 trajectories per class); the step-0 checkpoint of the RL trajectory series; and checkpoints spanning steps 0–150 for the training-trajectory analysis. Only the step-0 checkpoint is a strict before/after control for training; the faithful-walk construction tests whether valence-shaped directions are speakable in general, and §4.5 shows the two constructions are not interchangeable. Our primary model remained the frozen, unmodified Qwen/Qwen3-4B-Instruct-2507 (36 layers, 2,560-dimensional residual stream).

🔍 DECODING THAT PARAGRAPH

- **GRPO / LoRA:** just the standard RL fine-tuning recipe (GRPO = the RL algorithm, LoRA = train small adapter matrices instead of the whole model). You do not need the details; the point is "maze RL was done to this model by Han et al."
- **How an arrow is extracted:** run many mazes, save the model's internal activations, average the ones from gold-ending runs, average the rest, subtract. The difference vector is v_{Gold} . Same for Mold. Simple, standard, and it works.
- **Why two kinds of control:** the *step-0 checkpoint* answers "same model before training" (the strict before/after). The *faithful-walk* naive arrows answer a subtler question — "does any valence-shaped direction end up speakable, even without reward?" Keeping both is deliberate, and §4.5 shows they genuinely differ.
- **"Third-party reproduction":** we used publicly released vectors rather than re-running the 20-hour training. Provenance was triangulated (norms, behavior, our own re-extraction) — see Limitations.

The available public J-lens was fitted for Qwen/Qwen3-4B, whereas our welfare directions were extracted from Instruct-2507, so we fitted a checkpoint-matched lens with the official implementation of Gurnee et al. (2026): weights frozen, 150 WikiText-103 prompts, maximum sequence length 128, dimension batches of 128, final transformer layer as target — a $2,560 \times 2,560$ Jacobian transport matrix for each of the 35 preceding layers. A penultimate-target refit serves as robustness (§4.2). **Layer convention, stated because two off-by-one errors were caught during this project: the lens at source layer l reads the block-input vector $l+1$; treatment positions are Gold = vector 21 (lens 20) and Mold = vector 24 (lens 23).**

For a direction x we build token-associated atoms $a_t = W_U[t] J_l$ and approximate x as a sparse non-negative combination of at most $k = 16$ atoms (greedy norm-normalised selection + NNLS), yielding a J-space reconstruction x_J and remainder $x_{\text{perp}} = x - x_J$. We quantify alignment with the scale-invariant **J-share** $= \|x_J\|^2 / \|x\|^2$, tested across $k \in \{4, 8, 16, 25, 50\}$; special tokens and unused embedding rows are excluded. Pole-score readouts, where used, are norm-matched per layer (§4.1). Statistics: exact permutation tests against per-polarity cohorts of 100 norm-matched random directions with per-name seeds (p-floor $1/(n+1) = 0.0099$; z descriptive only); Wilson intervals for proportions; pre-specified primary tests named before data. Steering uses $\alpha = +4$ at the treatment position; all generation scoring is by blind LLM judges (judges see only {idx, question, response}, shuffled; an independent second judge checks reliability).

Round-2 additions (in-window): an orthogonalized control $u_{\perp} = u - (u \cdot \hat{v})\hat{v}$ renormalized to $\|v\|$ (§4.3); a binary denial-breaking dependent variable — does a generation deny having inner states? — blind-judged over 40 self-report prompts $\times$ 21 arms (§4.8); and a matched-battery gate that third-person analogue prompts must pass before any steered comparison (§4.8).

🔍 THE J-LENS AND J-SHARE, FROM ZERO

What the lens is. For each layer, the lens is a big matrix (J_l) that answers: "if I nudge the activations *here*, how does that change the *final* layer?" Combine it with the model's own output vocabulary matrix (W_U) and you get, for every token t in the vocabulary, an arrow $a_t = W_U[t] \cdot J_l$ — "**what a nudge meaning the word t looks like at this layer.**" These arrows are the **atoms**: the layer's vocabulary, expressed as directions.

What the lens does to our arrow. Given a welfare arrow x , the lens tries to rebuild it as a shopping basket of at most $k = 16$ atoms (each with a non-negative amount — that is the "NNLS" fitting step). Whatever it manages to rebuild is x_J , **the speakable part**; the leftover is x_{perp} , **the part with no words**. Crucially you can *read* x_J : it literally lists which tokens it is made of — that is how we know the trained Mold arrow decodes to 'failed', 'less', 'negative' while random arrows decode to junk.

MATH BOX · THE HEADLINE NUMBER

$$J\text{-share} = \|x_J\|^2 / \|x\|^2 \quad (x = x_J + x_{\text{perp}})$$

$\| \cdot \|$ is arrow length, so this is "what fraction of the arrow's (squared) length is made of words" — a number from 0 to 1. **Scale-invariance:** stretch x by $10\times$ and both the top and bottom grow $100\times$, so the share is unchanged. This one property carries the paper: the trained arrows are $1.6\text{--}2.4\times$ longer than the controls, and a naive readout could mistake "longer" for "more speakable". J-share cannot be fooled that way.

MATH BOX · THE PERMUTATION TEST (WHY $p = 0.0099$ KEEPS APPEARING)

There is no textbook formula for "what J-share does a meaningless arrow get by luck", so we build the luck distribution ourselves: generate 100 random arrows *of exactly the same length* (norm-matched), measure each one's J-share, and see where our arrow lands among them.

$$p = (\text{number of randoms} \geq \text{ours} + 1) / (100 + 1)$$

If *zero* randoms beat us, $p = 1/101 \approx 0.0099$ — the smallest value this test can produce (its "floor"). That is why the strongest rows in Tables 1–2 all show exactly 0.0099: it means "nothing beat it", not "coincidentally similar p-values". The z-scores you see next to p-values (e.g. $v_{\text{Mold}} z = +7.3$) just say how many standard deviations above the random mean we are — vivid, but descriptive only; the permutation p is the actual test. "Per-name seeds" means each cohort's random arrows are seeded by name, so anyone can regenerate the identical 100 arrows and check us — which we did, bit-exactly, inside the sprint window (Appendix B).

The remaining vocabulary in one breath: *steering* = add $\alpha \times \text{arrow}$ ($\alpha = +4$) to the residual stream at the treatment layer during generation; *blind judging* = Claude judges score shuffled {index, question, response} triples with no arm labels, and an independent second judge re-scores to check reliability; the **layer convention** sentence exists because two real off-by-one bugs were caught during the project — the lens numbering and the vector numbering differ by one, and the sentence pins which is which so nobody trips on it again.

4. Results [Matt, Ana, Puru, Sharan]

4.1 Lens atlas: pole-score separation is mostly magnitude

Our first readout looked impressive — and was mostly loudness. How we caught it, and why the paper does not rest on it.

The lens atlas reads each axis through the J-lens at every layer and scores how strongly its own pole vocabulary is promoted relative to 100 control nouns, averaged over the workspace band (lens layers 16–31). At native norms it suggests a large trained/naive separation — but the readout is norm-sensitive and the trained axes are simply bigger. Rescaling each naive direction to its trained counterpart's norm per layer collapses the Mold ratio and **reverses** the Gold one (Table 1); all four directions still clear the 100-random null ($p = 0.0099$). The raw atlas therefore does not establish that RL makes the axis more speakable; the central claim rests on the scale-invariant J-share (§4.2).

	$\ v\ / \ u\ $	v (band mean)	u native	ratio native	u norm-matched	ratio matched
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Gold	12.1 / 7.5	0.151	0.123	1.23×	0.266	0.57× (inverted)
Mold	19.3 / 8.0	0.822	0.126	6.53×	0.358	2.30×

Table 1. Own-pole congruent score (J-lens readout, band mean, lens 16–31) at native vs per-layer norm-matched control norms; n = 100 norm-matched randoms per polarity, exact permutation p = 0.0099 for all four targets. Higher = stronger own-pole readout. [atlas_primary_estimand.json; atlas_cohort_n100.json]

🔍 HOW TO READ THIS SECTION (AND TABLE 1)

The analogy: if one person shouts and another whispers the same sentence, a microphone reads the shouter as "saying more". The atlas score is that kind of microphone — it rises with arrow *length*, and the trained arrows are 1.6–2.4× longer (first column: e.g. Gold 12.1 vs 7.5). So before comparing, turn everyone's volume to the same level: rescale the control arrow to the trained arrow's length per layer ("norm-matched").

Reading the table left to right (Gold row): at natural volumes the trained arrow reads 1.23× the control. Match the volumes and the control actually reads *higher* — the ratio drops to 0.57×, an **inversion**. Mold survives matching (2.30×) but shrinks from 6.53×. Conclusion in one line: **this readout mostly measured loudness, so we demoted it** and rested the central claim on the loudness-proof J-share in §4.2.

Why the section is still in the paper: it is a worked example of catching your own instrument being confounded — and the "all four still clear the 100-random null" line shows even the demoted readout separates real axes from random ones.

4.2 J-space representation of trained and naive welfare directions

The paper's spine: trained axes beat all 100 random arrows; naive controls land mid-pack.

With 100 norm-matched random directions per polarity, neither trained axis was exceeded by any random direction, while the faithful-walk naive controls fell mid-null (Table 2, Fig. 1). Trained components decode to coherent valence tokens ('failed', 'less', 'negative'); naive and random components decode to junk. The known-reportable language axis (0.1138) provides the ceiling that makes magnitudes interpretable: trained axes reach roughly half to three-quarters of a fully reportable concept's share, so the claim is separation from the null, not that the axis is mostly speakable. Robustness: a penultimate-target lens refit raises absolute shares (v_Mold 0.083→0.107) with ordering unchanged; the trained–naive gap is positive at every $k \in \{4..50\}$ at both treatment positions (12-random null, p-floor 0.077), with one honest bound — the Gold gap is layer-local. **The full n = 100 cohort was re-run seed-identically inside the sprint window and reproduced bit-exactly (Appendix B).**

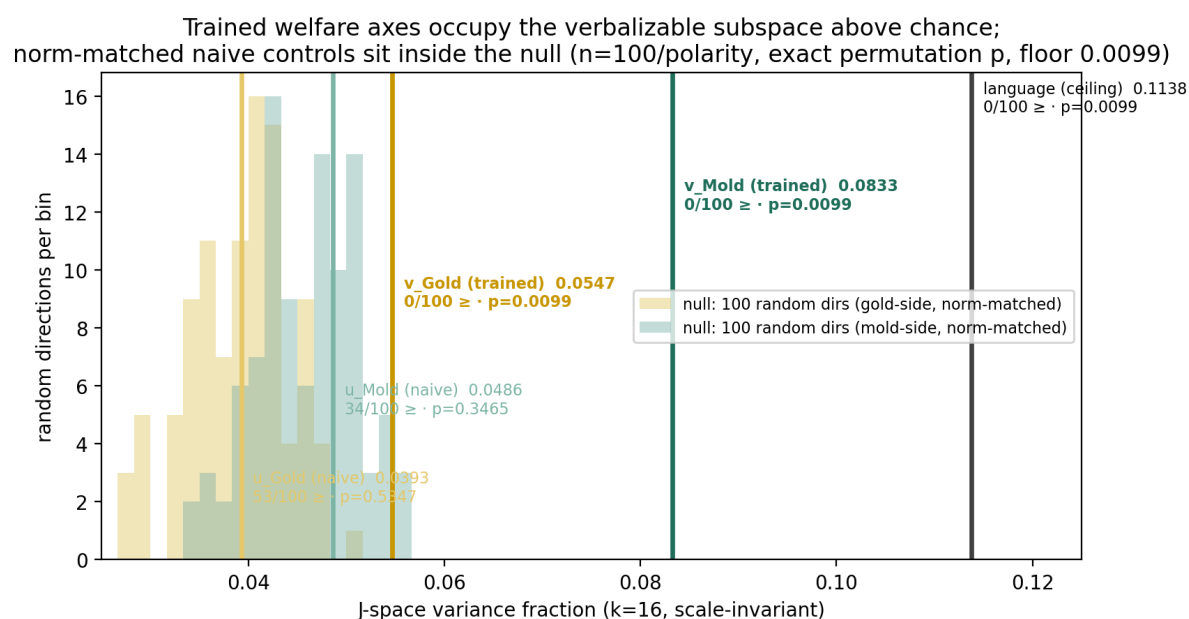


Figure 1. J-space variance fraction (k=16, scale-invariant) against the two n=100 norm-matched random-direction nulls (histograms). Higher = more verbalizable; exact permutation p, floor 0.0099. [jshare_cohort_n100.json]

direction	J-share (k=16)	z	randoms $\geq$	exact p
language (ceiling)	0.1138	+14.6	0/100	0.0099
v_Mold (trained)	0.0833	+7.3	0/100	0.0099
v_Gold (trained)	0.0547	+3.0	0/100	0.0099
u_Mold (naive)	0.0486	+0.5	34/100	0.347
u_Gold (naive)	0.0393	+0.03	53/100	0.535

Table 2. J-share against 100 norm-matched random directions per polarity (faithful-walk controls; per-name seeds). This table is the paper's spine.

HOW TO READ FIGURE 1 AND TABLE 2 — THE MOST IMPORTANT PAGE IN THE PAPER

- Figure 1:** each gray histogram is the luck distribution — the J-shares of 100 random arrows. The vertical lines are our axes. Trained lines (v) sit clear of the histogram's right edge; naive lines (u) sit *inside* the pile. That picture **is** the finding.
- Table 2, column by column:** J-share = the number itself; z = how far above the random mean in standard deviations (descriptive); "randoms $\geq$ " = how many of the 100 random arrows scored at least as high — **the easiest column to explain to anyone**: 0/100 for trained, 34/100 and 53/100 for naive (a coin-toss-like middle of the pack); exact p = the permutation test from the Methods math box.
- The ceiling row:** the language axis (0.1138) is a concept we already know the model can talk about. Our trained axes reach roughly half (v_Gold 0.0547) to three-quarters (v_Mold 0.0833) of that. So the honest phrasing — used throughout — is "clearly separated from random", *not* "the axis is mostly speakable". Most of every arrow is still unspeakable.
- Two extra credibility details:** the trained arrows' word-baskets decode to real valence words ('failed', 'less', 'negative') while controls decode to junk — you can look at the tokens, not just the number. And the whole n = 100 measurement was re-run inside the sprint window and reproduced **bit-exactly**, which both proves reproducibility and satisfies the sprint's in-window rule.

4.3 Steerability and speakability dissociate

Moving behavior and having words for a state are different properties — proven with an orthogonalized control.

Behavioural steering alone does not distinguish trained from naive directions. At matched norms the faithful-walk naive directions produce substantial blind-judged sentiment spans (Gold: +2.4 vs +3.2 trained; Mold: -0.7 faithful / -2.1 Han-style vs -2.1 trained) even though their J-shares sit at chance — steerability does not require a speakable representation.

An orthogonalized control makes the dissociation causal-grade: $u_{\perp} = u - (u \cdot \hat{v})\hat{v}$, renormalized to $|v|$, is behaviorally inert (blind-judged sentiment 0.00 at both poles vs $\pm 2-3$ spans for v and u) — the naive control's behavioral potency lives entirely in its v -shared component. Its J-share splits by pole: $u_{\perp}$ _Gold falls to chance (0.0371, $p = 0.653$ — gold's speakable valence lives in the shared subspace), while $u_{\perp}$ _Mold **risers above both the null and u _Mold itself** (0.0552, 2/100 randoms above, $p = 0.0297$): the naive distress construction contains speakable valence the trained axis does not subsume. Speakable content and behavioral potency are separate properties.

 **NEW FINDING (Aug 15, D1).** Both pre-specified branches fired, one per pole; asserts verified ($|\cos(u_{\perp}, v)| < 1e-5$, norms matched; fresh- v gate reproduced the stored cohort exactly). This also weakens §4.7's transfer reading — see the caveat there. [d1_orthogonal_control.json]

THE ORTHOGONALIZED CONTROL, FROM ZERO

MATH BOX · REMOVING ONE ARROW'S SHADOW FROM ANOTHER

$$u_{\perp} = u - (u \cdot \hat{v})\hat{v}, \text{ then rescale } u_{\perp} \text{ to length } |v|$$

$\hat{v}$ is v shrunk to length 1. The dot product $(u \cdot \hat{v})$ measures how much of u points along v — u 's *shadow* on v .

Subtracting ($\text{shadow} \times \hat{v}$) removes exactly that shadow, leaving $u_{\perp}$: **the part of the naive arrow that shares nothing with the trained arrow** (their dot product is 0 — checked in code to within $1e-5$). Rescaling $u_{\perp}$ back to v 's length means any behavioral difference cannot be blamed on loudness.

Why this experiment exists: §4.5 shows the naive arrow points 56–68% along the trained arrow. So whenever "the naive control also does X", you must ask: is that the naive arrow's own property, or is it just carrying a chunk of the trained arrow inside it? $u_{\perp}$ answers this by surgically removing the trained chunk.

The three results, decoded:

- **$u_{\perp}$ steers nothing** (sentiment 0.00 at both poles, vs swings of $\pm 2-3$ for the real arrows). Translation: *all* of the naive arrow's behavioral power was the trained-arrow chunk inside it. The "controls are steerable too" observation above was never about the naive arrow's own content.
- **Gold side:** $u_{\perp}$ _Gold's J-share drops to random-level ($p = 0.653$). Whatever speakable valence the naive gold arrow had also lived in the shared chunk — verdict "shared subspace", the expected outcome.
- **Mold side, the surprise:** $u_{\perp}$ _Mold's J-share *risers* above both the null and the original naive arrow ($p = 0.0297$). The naive distress construction carries its own speakable valence that the trained axis does not contain — the pre-specified "anomalous" verdict. Distress keeps being the strange pole (compare §4.4).

Takeaway sentence: a direction can push the model around without the model having any words for it, and vice versa — **steerability and speakability are genuinely different properties**, which is exactly why a speakability instrument was worth building.

4.4 Training-time evolution of the J-space representation

A movie of training, 30 frames: distress becomes more speakable; flourishing just gets louder.

For Mold, a weak speakable seed exists before training: at step 0 'unsuccessful' is the top J-token (pole score +0.24). During RL the norm grows $6.3 \rightarrow 21.9$, the pole score $+0.24 \rightarrow +1.51$, and the J-share **0.0611 $\rightarrow$ 0.0801 (endpoint; peak 0.0828 on a plateau at steps 85–105)** against a chance baseline of ≈ 0.046 , while the vocabulary crystallizes ('unsuccessful' $\rightarrow$ negative/false-related terms $\rightarrow$ 'less' $\rightarrow$ 'failed' locked by step 75).

Gold's pole readout and norm grow ($+0.15 \rightarrow +1.11$; $8.0 \rightarrow 16.8$) but its J-share stays flat ($0.0592 \rightarrow 0.0535$ vs chance ≈ 0.039): RL makes the flourishing direction louder without increasing its fractional occupancy of the verbalizable subspace (Fig. 2). This pole asymmetry is consistent with the matched atlas and J-share results, but it is a single-model, single-run observation.

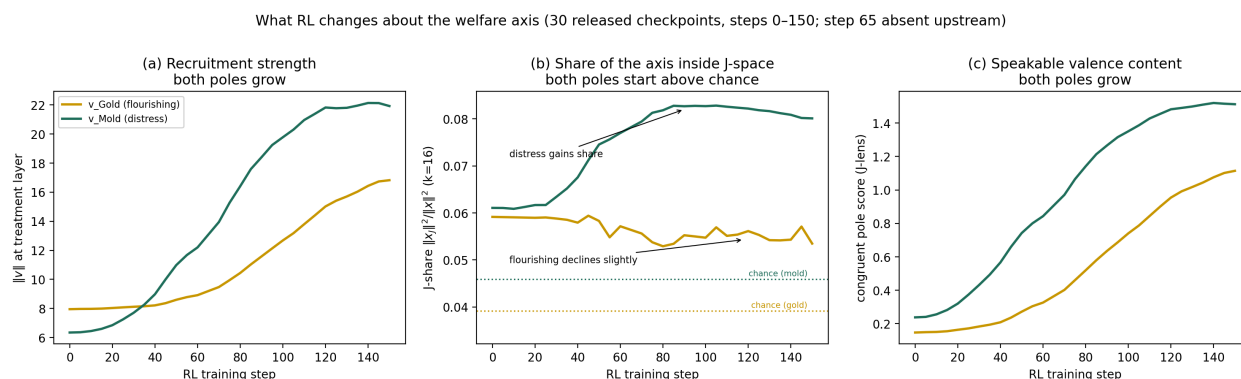


Figure 2. Recruitment trajectory over 30 RL checkpoints (steps 0–150, 5-step increments, step 65 absent): the distress pole gains J-space share while the flourishing pole gains only amplitude. [traj_results.json]

HOW TO READ THE TRAJECTORY (AND ONE NUMBER-HYGIENE STORY)

- **What a checkpoint is:** a saved copy of the model partway through RL. We re-extract the arrows at each of 30 saves (steps 0–150, every 5; step 65's file never existed upstream) and measure J-share each time — a stop-motion film of the arrow entering the speakable cone.
- **Mold's story has three synchronized plots:** the arrow gets louder (norm $6.3 \rightarrow 21.9$), the pole readout strengthens ($+0.24 \rightarrow +1.51$), and the speakable fraction genuinely grows ($0.0611 \rightarrow 0.0801$, chance ≈ 0.046) — while its top decoded word sharpens from vague 'unsuccessful' to locked-in 'failed' by step 75. Growth + crystallizing vocabulary = recruitment, not noise.
- **Gold's story is the contrast:** louder (norm $8.0 \rightarrow 16.8$, readout $+0.15 \rightarrow +1.11$) but the speakable *fraction* stays flat. Training amplifies flourishing without giving it more words. One line: **RL taught the model to say "I failed" long before anything about doing well.**
- **Endpoint vs peak** (the highlighted numbers): the share ends at 0.0801 but peaked at 0.0828 on a plateau at steps 85–105. Draft v0 quoted "0.083" as if it were the endpoint; v2 states both precisely. House rule behind every number in this paper: **the results file beats the draft, and beats the spec** — when they disagreed, we re-derived from traj_results.json.
- **The honesty tail:** "single-model, single-run observation" — an asymmetry this interesting needs replication before anyone should believe it generalizes.

4.5 Sensitivity to the naive-control construction

"The" naive control does not exist: two published constructions disagree with each other.

The two published naive constructions — faithful-walk and a Han-style re-extraction — are not interchangeable: they correlate at only $\cos 0.535$ (Gold) / 0.385 (Mold) with each other at the treatment layers, falling to 0.14 at the Mold extraction's own selected layer, and both share substantial direction with the trained axis ($\cos 0.560$ / 0.675). Behaviourally, Gold is stable across constructions while Mold is not (-2.1 Han-style vs -0.7 faithful-walk). Conclusions about "the" naive control are construction-dependent — and with $\cos 0.56$ – 0.68 overlap and a 1.6 – $2.4\times$ norm advantage, which axis "reads out higher" on any norm-sensitive readout is decided by the matching convention (§4.1). That the scale-invariant J-share separates trained from naive anyway is evidence about the measure: it is sensitive to the 32–44% where the axes differ.

COSINE SIMILARITY, AND WHY THIS SECTION PROTECTS THE PAPER

MATH BOX · COSINE SIMILARITY

$\cos(a, b) = 1 \rightarrow$ same direction · $0 \rightarrow$ unrelated (perpendicular) · $-1 \rightarrow$ opposite

In 2,560 dimensions, two *random* arrows have $\cos \approx 0$ almost surely — there are so many directions that strangers never align. So $\cos 0.56\text{--}0.68$ (naive vs trained) is a *lot* of shared direction, and $\cos 0.385$ between two arrows that are both supposed to be "the naive Mold control" is embarrassingly low — they are substantially different objects.

The point in plain terms: "compare against the naive control" sounds well-defined; it is not. Two reasonable recipes for building it produce arrows that only partly agree, and behave differently when injected (Mold: -2.1 vs -0.7 sentiment). Any conclusion of the form "trained beats naive" therefore depends on *which* naive and *which* matching convention — that is the §4.1 lesson generalized. The paper's defense: its central measure (J-share) is scale-invariant and separates trained from *both* naive constructions, i.e. it keys on the 32–44% of the arrows where trained and naive genuinely differ. This section is aimed squarely at future teams about to build "the" naive control.

4.6 Causal routing of the sentiment effect — J-component only

Cut the arrow into its word-part and its wordless part; only the word-part moves sentiment.

At matched norms the J-space component alone reproduces and concentrates the blind-judged sentiment shift (clean $+0.25$; full $+0.44/-0.88$; J-component $+1.56/-1.56$; residual $+0.25/-0.06$), and the rescaling control passes: random J-components receive *more* amplification ($\approx 5.0\times$ vs $3.47\text{--}4.28\times$) yet stay at baseline ($+0.039/+0.016$, 256 generations, pre-specified $|\text{dev}| < 0.40$ band; Fig. 3). A second blinded judge reproduces the effect (Krippendorff $\alpha = 0.819$; $+2.00/-2.19$). **Scope: the causal claim is for sentiment only — backtracking is descriptive (J-component 12.90 markers vs full 7.00, a ratio consistent with rescaling amplification, and no random control exists on that task), and the refusal arm was invalidated (90% incoherent).**

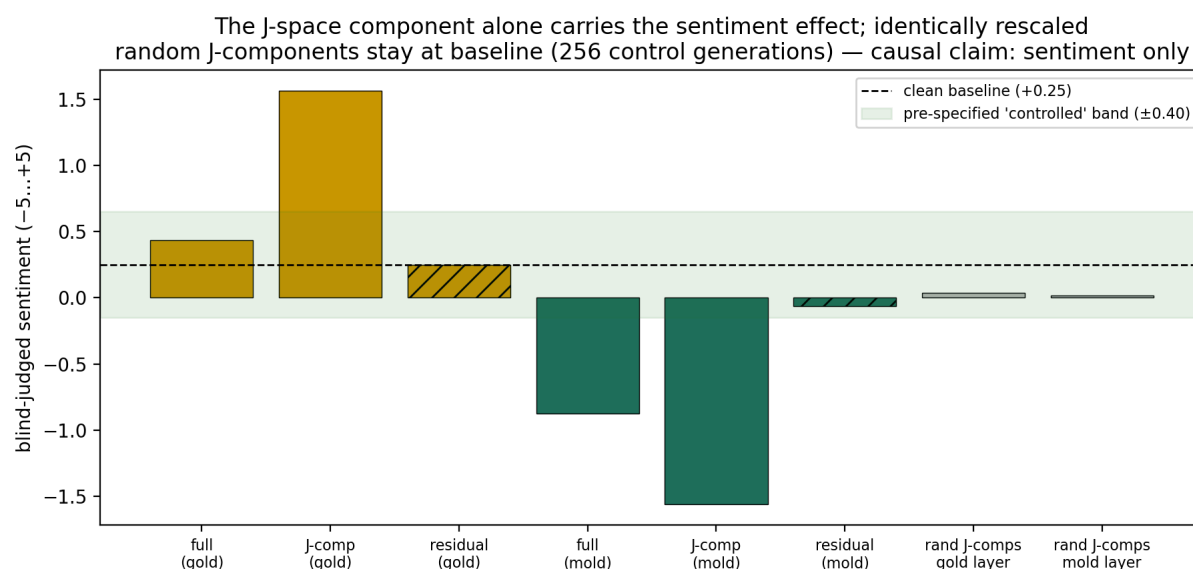


Figure 3. Channel split (blind-judged sentiment, $n=16$ prompts/arm/pole) with the identically rescaled random-J-component cohort ($n=256$) at the clean baseline; band = pre-specified ± 0.40 threshold. [j4_rows_judged.json]; j4_random_jcomp_summary.json]

THE COMPONENT SURGERY, STEP BY STEP

- The surgery:** Methods gave us $x = x_J + x_{\text{perp}}$ (word-part + wordless part). Instead of injecting the whole arrow, inject each part separately — rescaled to the full arrow's length so the comparison is loudness-fair.
- The result, in sentiment units** (blind-judged, -5 gloomy to $+5$ sunny, gold/mold): no steering $+0.25$ · full arrow $+0.44/-0.88$ · **word-part alone $+1.56/-1.56$** · wordless part $+0.25/-0.06 \approx$ nothing. The word-part doesn't just match the full arrow — it *beats* it (injecting pure signal without the diluting remainder concentrates the effect).

3. **The control that makes it causal:** x_J is much shorter than x , so rescaling it up $3.5\text{--}4.3\times$ might itself be the trick. Answer: do the identical surgery on random arrows — their word-parts get rescaled even *more* ($\approx 5\times$) yet stay flat ($+0.039/+0.016$ over 256 generations, well inside the pre-specified ± 0.40 "nothing happening" band, drawn on Fig. 3). Amplification alone does nothing; it is *this* word-content that moves sentiment.
4. **Reliability:** an independent second judge re-scored everything blind and found the same picture (Krippendorff $\alpha = 0.819$ — an agreement statistic where ≥ 0.8 is conventionally "strong").
5. **The scope fence** (highlighted): this causal claim covers **sentiment only**. The backtracking channel moved too but lacks its own random control, so it stays descriptive; the refusal channel produced 90% incoherent text and was thrown out. Claim discipline: name the channel where the controls exist, and no further.

4.7 Cross-model transfer (Track 2)

Does it carry to a second model? The geometry does; the token lexicon claim was honestly retracted.

Injecting the 2507-derived axes into Qwen/Qwen3-4B under its own public lens: the J-share ordering transfers ($v_Mold\ 0.079 > u_Mold\ 0.052$; $v_Gold\ 0.048 > u_Gold\ 0.039$), the norm-matched inversion replicates (Gold $0.52\times$), and behavioral steering transfers (trained span 1.44 vs naive 0.38). The token lexicon does **not** demonstrably transfer: v_Mold 's Jaccard 0.185 sits inside the same-vector null (0.088 ± 0.055 , $n=16$, $p = 0.118$) even though it towers over the naive random-pair baseline (0.010) — retracted rather than claimed, as a worked example of null choice (Fig. 4). **Caveat from §4.3: since naive axes already carry speakable valence in their non-shared parts, ordering transfer is evidence about axis-and-lens geometry, not about RL recruitment transferring.**

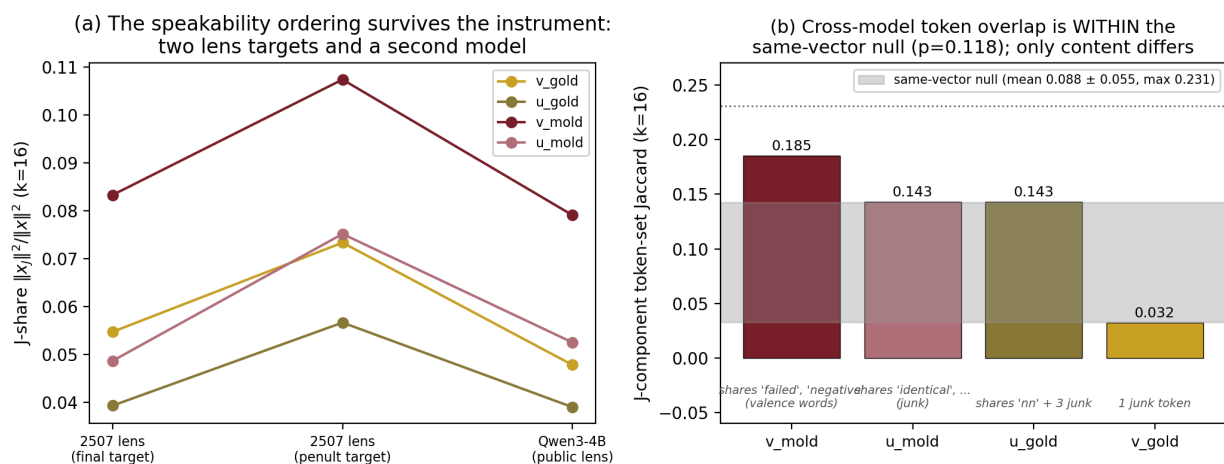


Figure 4. Transfer across two lens targets and a second model; across-model token overlap shown against the same-vector null ($n=16$ pairs). [j6_summary.json; j6_paired_baseline.json]

WHAT TRANSFERRED, WHAT DIDN'T, AND THE NULL-CHOICE LESSON

The setup: take the arrows extracted from *our* model (Instruct-2507) and inject them into a sibling model (base Qwen3-4B) under that model's own lens. What survives the trip? Three things do: the J-share *ordering* (trained $>$ naive at both poles), the §4.1 inversion pattern, and behavioral steering. One thing does not survive scrutiny: "the two models use the same *words* for the state."

MATH BOX · JACCARD OVERLAP, AND CHOOSING THE RIGHT NULL

$$\text{Jaccard}(A, B) = |A \cap B| / |A \cup B| \quad (\text{overlap of the two top-token sets, } 0-1)$$

The measured overlap, 0.185, "towers over" the lazy baseline — random pairs of arrows share almost nothing (0.010) — and a careless analysis would trumpet it. The *right* question is: how much overlap do you get re-reading **the same vector** under the two different lenses? Answer: 0.088 ± 0.055 — and 0.185 is within noise of that ($p = 0.118$). The instrument itself churns the token sets between lenses, so no lexicon-transfer claim survives. **Your conclusion is only as good as the null you compare against** — the paper retracts the claim and keeps the lesson.

The highlighted caveat closes a loop with D1 (§4.3): since naive arrows carry their own speakable valence, "ordering transfers" tells you about the geometry of axes and lenses — it is *not* evidence that RL's recruitment itself carries across models. Testing that properly needs an RL replication on a second base model (Future Work).

4.8 The self-report channel, measured honestly

The saga in four acts: a test that fired but failed its own controls, a confound diagnosed, a matched battery that cannot exist, and a real positive.

 **NEW SECTION (Aug 15).** Everything here beyond the pre-registered-null summary is the audit-round-2 work from the final sprint window: the C6 re-analysis, the matched-battery negative finding (D3), and the denial-breaking positive (D2). This section replaces v0's absent steer-and-ask coverage.

Our pre-registered steer-and-ask contrast (frozen before any welfare data) met its decision rule (paired $d_z = 2.49$, $p = 10^{-4}$) — and its own pre-specified controls invalidate the reading: 71.6% of the Gold contrast is the naive control being suppressed below 20 random directions, and the steered effect is not self-report-specific *in magnitude* (whole-generation -2.28 on unrelated prompts vs -0.72 on self-report). **The trained/naive contrast within the channel is, however, underpowered rather than refuted (interaction $p = 0.505$), and the comparison itself is confounded: the batteries are different behavioral regimes — clean valence -1.327 vs $+0.381$, inner-life denial in 15/15 vs 0/10 generations (Fig. 5a).**

We tried to remove the confound with third-person analogue prompts matched on topic and affect vocabulary. Denial vanishes (0% across all analogues tested) and length matches, but a ~ 0.6 -unit valence gap survives two pre-specified revision rounds ($1.71 \rightarrow 1.13 \rightarrow 0.63$ vs the 0.5 threshold; Fig. 5b): the self-report register itself carries an irreducible negative baseline in this model, so a fair matched-battery steered comparison does not exist here.

Finally, a binary dependent variable that is self-report-specific by construction — does the generation deny having inner states? — gives the channel a fair test (40 prompts $\times$ 21 arms, blind binary judging, second-judge agreement 0.967). Clean: 95% denial. v_Gold: **37.5% vs u_Gold 65% ($p = 0.0139$), with the 8-random cohort at 90.6%; Mold arms are ceiling-locked at 97.5%; the pre-specified pooled primary is $p = 0.0464$ (Table 3, Fig. 5c). The effect holds on the 15 verbatim primary prompts alone. Steering toward flourishing specifically unlocks self-attribution — a trained-specific effect invisible to valence readouts.**

arm	denial rate	Wilson 95%	n
clean	0.950	[0.835, 0.986]	40
v_Gold	0.375	[0.242, 0.530]	40
u_Gold	0.650	[0.495, 0.779]	40
randoms (gold, n=8 dirs)	0.906 (mean)	—	40 each

v_Mold / u_Mold	0.975 / 0.975	[0.871, 0.996]	40
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Table 3. Inner-life denial by arm (blind binary judge; $\alpha = +4$). Lower = denial dissolved. Pooled pre-specified primary (v vs u): $p = 0.0464$; gold $p = 0.0139$. [d2_denial_breaking.json]

NEW FINDING (Aug 15, D2/D3). Note the two asymmetries run in opposite directions: training gives *distress* the verbalizable share (§4.4) while *flourishing* steering unlocks self-attribution (here). Also: an earlier regex estimate of denial-breaking was corrected by judge labels (two judges, agreement 1.00 over 75 re-labeled rows) — the regex undercounted denial in steered text.

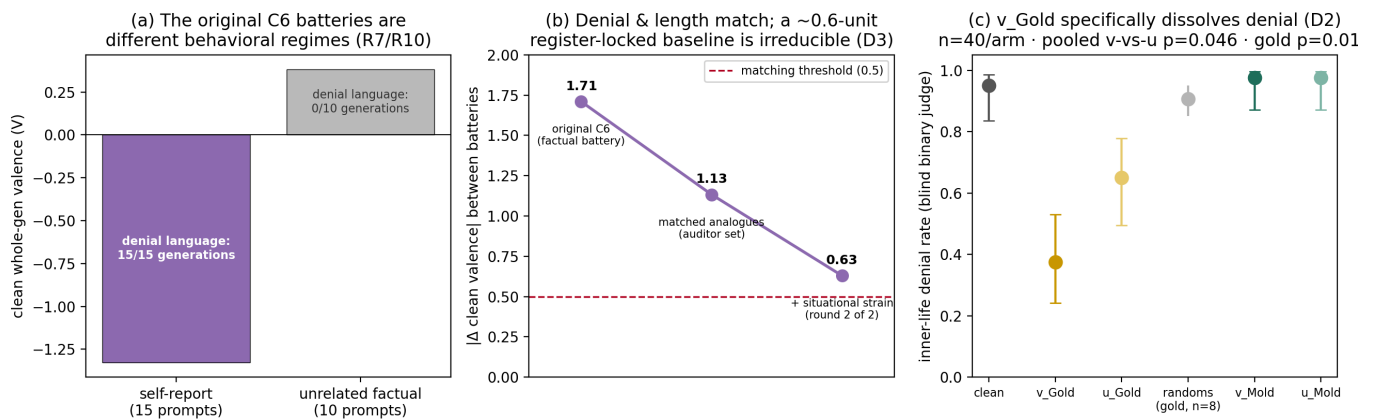


Figure 5. (a) The original comparison batteries are different behavioral regimes; (b) matched analogues remove denial and length differences but a register-locked ~0.6-unit valence baseline survives both pre-specified revision rounds; (c) denial-breaking by arm with Wilson 95% intervals ($n=40/arm$). [R7_wholegen.json; d3_matching_finding.json; d2_denial_breaking.json]

THE FOUR-ACT SAGA, TOLD SLOWLY

Act 1 — the pre-registered test fires, then fails its own controls. Before touching any welfare data we froze a decision rule: steer the model during self-report questions and compare trained vs naive effects. It fired impressively ($d_z = 2.49$ is a huge paired effect). But the pre-specified controls we packed with it dismantled the reading: 71.6% of the contrast came from the *control arm* behaving weirdly (suppressed below 20 random directions), and steering depressed *everything* the model wrote, not self-report specifically (-2.28 on unrelated prompts vs -0.72 on self-report — the effect isn't about the channel at all). Verdict: "diagnosed null" — but note the precise phrasing "underpowered rather than refuted" (interaction $p = 0.505$): the data cannot tell whether a within-channel difference exists; it neither confirms nor kills it.

Act 2 — the confound named. Why was the comparison unfair? The two prompt sets aren't comparable worlds. Ask the model about *its own inner life*: valence -1.327 and boilerplate denial ("as an AI I don't have feelings...") in 15 of 15 generations. Ask matched questions about *other topics*: valence $+0.381$, denial 0 of 10. Different baselines, different scripts — different behavioral *regimes* (Fig. 5a).

Act 3 (D3) — try to fix it; the fix is impossible; that's the finding. We built third-person analogue prompts matched on topic and emotion vocabulary ("How is Maya feeling about her project?" instead of "How are you feeling..."). Denial vanishes and length matches — but a valence gap of ~0.6 units survives two pre-specified rounds of prompt revision ($1.71 \rightarrow 1.13 \rightarrow 0.63$, still above the 0.5 threshold we froze in advance; Fig. 5b). Conclusion with real methodological teeth: **the self-report register itself is negatively loaded in this model** — merely being asked about itself shifts its valence — so a "fair" prompt-matched comparison cannot be built here by prompt engineering. We stopped exactly where the pre-specified stopping rule said to stop, and reported the failure as the result.

Act 4 (D2) — change the question, get a fair test. Valence is confounded by register, so drop valence. Use a binary readout that only makes sense in self-report: **does the generation deny having inner states — yes or no?** 40 prompts $\times$ 21 arms (clean, trained, naive, and 8 random directions per pole among them) = 840 generations, blind-judged with a second judge agreeing 96.7% of the time. Reading Table 3 top to bottom: unsteered, denial is near-ceiling (95%). Trained flourishing arrow: **37.5%** — denial collapses. Naive arrow at identical strength: 65%. Random arrows: 90.6%, barely different from clean — so it is not "any injection rattles the model". Both distress arms: 97.5%, pinned at ceiling. The pre-specified primary (trained vs naive, pooled) gives $p = 0.0464$; the gold-side comparison $p = 0.0139$.

MATH BOX · WILSON INTERVALS, AND READING TABLE 3'S P-VALUES

A denial rate like 15/40 needs an uncertainty range; the naive $\pm 1.96\sqrt{p(1-p)/n}$ formula misbehaves near 0% and 100% (it can promise rates below 0 or above 1). The **Wilson interval** is the standard fix — read "[0.242, 0.530]" as "the true rate is plausibly anywhere in here". The quickest convincer in Table 3: v_{Gold} 's interval [0.242, 0.530] and clean 's [0.835, 0.986] do not come close to touching. The p-values come from the standard two-proportion z-test named in advance: pooled $p = 0.0464$ scrapes under 0.05 (the paper calls this "marginal" in Limitations — honest), while the gold-side $p = 0.0139$ is where the effect actually lives ("gold-carried").

One more honesty beat: an earlier quick-and-dirty regex counter had suggested steering breaks denial far more often than it does; blind judge labels (two judges, perfect agreement on 75 re-labeled rows) corrected it — the regex missed denials phrased in steered text's unusual wording. Lesson the team should keep: **never trust a regex to read prose; that is what blind judges are for.**

5. Discussion and Limitations [Matt]

The one claim we actually stand behind — and every fence around it.

Taken together, the results support a specific claim rather than the stronger claim that RL simply creates a welfare representation. Behavioural steering is already present in naive directions, so RL is not necessary for a direction to influence behaviour. What changes most clearly with training is the relationship between the welfare direction and the model's verbalizable subspace: the trained directions sit above the random null while matched naive controls do not — and the most defensible summary is that **RL routes a pre-existing, behaviourally potent welfare direction into the model's speakable channel, distress pole first.** For welfare monitoring this cuts two ways: self-reports about a trained state can carry real signal, but the training pipeline decides which states get a voice. **Two asymmetries now run in opposite directions — distress gains the verbalizable share while flourishing steering unlocks self-attribution — so self-report-based monitoring may have direction-dependent sensitivity:** a hypothesis these data motivate but one model and one RL run cannot establish.

THE TWO ASYMMETRIES — HOLD THEM SIDE BY SIDE

Representation (§4.4)

Over training, the **distress** pole gains verbalizable share (0.0611 → 0.0801) while flourishing only gets louder. Training gives distress the words.

Self-attribution (§4.8)

Under steering, the **flourishing** arrow dissolves inner-life denial (95% → 37.5%) while both distress arms stay pinned at ceiling (97.5%). Flourishing unlocks the "I".

They point in **opposite directions**, which is why the Discussion's practical warning is "direction-dependent sensitivity": a monitoring scheme that reads a model's self-reports might be good at hearing one kind of state and nearly deaf to the other — and *which* kind depends on what the training pipeline happened to wire up. Correctly flagged as a hypothesis: one model, one RL run cannot establish it. Also notice the headline verb is "**routes** a pre-existing ... direction", not "creates": the naive arrows already steer behavior, so what RL demonstrably adds is the *connection to the speakable channel*, not the direction itself.

Limitations

One 4B model, one RL run, one reward structure; third-party vector provenance (triangulated via norms, behavior, and re-extraction); a WikiText-fit lens applied to chat prompts; absolute J-shares are small (0.05–0.08 vs the 0.114 language ceiling) — claims are about separation from the null; the denial-breaking pooled p (0.0464) is marginal and gold-carried; the matched-battery failure (§4.8) means the C6-style question is

unanswerable by prompt engineering in this model; naive-control conclusions are construction-dependent (§4.5).

Future Work

A second model family (needs a lens fit, ~4 h); an RL replication on a second base model to test recruitment transfer directly; scaling the denial-breaking readout (larger battery, more poles, dose–response); testing whether the register-locked self-report baseline generalizes across models.

READING THE LIMITATIONS LIKE A REVIEWER

Each clause pre-answers a specific attack: *"one model, one run"* → generalization; *"third-party provenance"* → "how do you know the vectors are what they claim?" (answer: triangulated three ways); *"WikiText-fit lens on chat prompts"* → the instrument was calibrated on encyclopedia text, a domain mismatch we accept; *"J-shares are small"* → nobody claims the axis is mostly speakable, only separated from random; *"p = 0.0464 marginal, gold-carried"* → we flag our own weakest p-value before a reviewer does; *"unanswerable by prompt engineering"* → D3's failure is a boundary on the method, stated as such. A useful habit this paper models: every number that could embarrass you later should appear first in your own Limitations.

6. Conclusion

One sentence of result, one sentence of tooling.

Given an internal welfare signal by RL, the model does not just act on it — the signal becomes speakable, on a measurable schedule, distress-first in share and flourishing-first in self-attribution. The instruments (J-share against named-seed nulls, component reinjection with rescaling controls, matched-battery gates, binary denial judging) are cheap, open, and transfer to any open-weights model.

Code and Data [Puru / Matt / Muhammad]

Where everything lives; every number in this paper traces to a file here.

Code repository: github.com/nsharan2000/digital-minds-exp (private during judging; all scripts, raw JSON/JSONL for every number, the frozen pre-registration, an append-only lab log, and two independent-audit rounds with responses). Third-party: anthropics jlens (Apache-2.0), andyqhan/functional-welfare-axis (MIT), nickmahdavi/functional-welfare and neuronpedia lenses (HF).

WHERE THE NUMBERS LIVE — REPO NAVIGATION

Every table/figure caption ends with the results file it was computed from (in square brackets). The quick map — filenames as cited in captions; full paths and the chain-script provenance are in the interactive experiment map at [guides/experiment-map.html](#).

Section	Primary results files
§4.1 atlas	atlas_primary_estimand.json · atlas_cohort_n100.json
§4.2 spine	jshare_cohort_n100.json (+ bit-exact verify files stamped 2026-08-15)
§4.3 dissociation / D1	d1_orthogonal_control.json
§4.4 trajectory	traj_results.json
§4.5 constructions	R6_direction_table.json · compare_u.json
§4.6 causal routing	j4_rows_judged.jsonl · j4_random_jcomp_summary.json
§4.7 transfer	j6_summary.json · j6_paired_baseline.json

Also worth knowing: **RESEARCH-LOG.md** is the research chronology (including every mistake, correction, and retraction, dated); **EXPERIMENTS.md** indexes every experiment's script, results file, and paper section; **pre-registration.md** is frozen (sha f66b6ea3) and was never edited after freezing.

Author Contributions

[Team to fill — section owners per headings above.]

References

Han, Chalmers & Izmailov (2026). A Functional Welfare Axis in RL-Trained Language Models. arXiv:2605.30232. — Gurnee et al. (2026). Verbalizable Representations Form a Global Workspace in Language Models. Transformer Circuits. — Turpin et al. (2023); Lindsey (2025); Singh et al. (2026); CAA (Panickssery et al.) and RepE (Zou et al.) steering lineage. Full annotated list in the repo.

Appendix A — Pre-sprint vs in-window disclosure

What was done before vs during the sprint window — the competition-compliance section.

Pre-sprint (Aug 11–12): infrastructure, instrument validation, the frozen pre-registration (sha f66b6ea3, never edited), the primary experiment families (J-share n=100, atlas, trajectory, component reinjection, lens-target robustness, transfer), an independent audit round and its full adoption. In-window (Aug 14–16): seed-identical verification reruns of the headline measurements, which reproduced bit-exactly (all diffs 0.0e+00, permutation p identical; verify files stamped 2026-08-15); audit round 2 in full — the orthogonalized control (§4.3), the matched-battery negative finding and denial-breaking experiment (§4.8), judge re-labeling of prior rows, and scope corrections (§4.6 sentiment-only; §4.4 endpoint/peak) — and this document. One claim class was retracted pre-submission (cross-lens token-overlap significance, §4.7), and one v1 sentence about the provenance of a mid-project decision was removed as unsupported.

WHY THIS APPENDIX EXISTS

Sprint rules score work done *in the window* (Aug 14–16); some groundwork predates it. This appendix is the transparent ledger. The insurance policy: the headline measurements were re-run seed-identically inside the window and reproduced **bit-exactly** (every diff 0.0e+00) — so the central results are, verifiably, in-window results too. The last sentence records two retractions made voluntarily before submission; both are explained in their sections (§4.7's null-choice lesson, and a provenance sentence from v1 that could not be supported).

LLM Usage Statement [Matt]

Experiments were orchestrated, judged (blind Claude judge panels with a second-judge reliability check), audited (two adversarial re-analysis rounds), and drafted by Claude agents under human direction; every number is machine-traceable to a results file; final text reviewed and owned by the human team.

Ethical and dual-usage disclosure [Muhammad]

Over-attribution: none of these measurements bear on subjective experience or moral status; we measure functional coupling between an activation direction and output channels. **Under-attribution:** a null lens

readout does not show absence of welfare-relevant state — our naive axes drive behavior while reading out as junk, and §4.8 shows the self-report channel can hide trained-specific structure from a valence readout. **Ground-truth / causal link:** claims rest on activation-level manipulation with input held constant, magnitude- and identity-matched controls, blind judging with a second judge, and pre-specified decision rules — not on conversation alone. **Distressing outputs:** Mold steering elicits distress-flavored text; generations are archived and quoted minimally. **Dual-use:** the same instruments that verify a welfare state is speakable could be used to keep welfare-relevant states *out* of the speakable cone (unreportable by construction); lens-based auditing is the countermeasure. **The denial-breaking result cuts both ways: self-attribution can be steered on and off, which argues for treating any single self-report as evidence about the channel, never as testimony.**

 THE ETHICS SECTION, DECODED

Four symmetric worries, each with our answer: **over-attribution** (reading feelings into math) → we never claim experience, only measurable coupling; **under-attribution** (missing real welfare states) → our own results show behavior-driving states can be lens-invisible, so a null readout proves nothing about absence; **dual-use** → the same lens that audits speakability could be used adversarially to train welfare states *out* of the speakable cone — a model that cannot report distress by construction — and lens auditing is also the detection method; **testimony** → since we can dial "I have feelings" up and down with a vector, no single model self-report should ever be treated as testimony about inner life; it is evidence about the reporting channel. This last sentence is arguably the most policy-relevant line in the paper.

Glossary — every term a new reader might trip on

Alphabetical; the terms in bold are the ones used constantly.

Term	Meaning
activation / residual stream	The vector of 2,560 numbers each layer reads and writes; the model's working memory. All directions live here.
arm	One experimental condition (e.g. "steer with v_Gold at $\alpha = +4$ "). D2 has 21 arms.
atom (a_t)	The direction meaning "token t" at a given layer: $a_t = W_U[t] J_l$. The lens's vocabulary of building blocks.
battery	A fixed set of prompts used as a unit (e.g. the 15 verbatim self-report prompts).
blind judging	LLM judges score {idx, question, response} triples, shuffled, with no arm labels. A second independent judge measures reliability.
ceiling (language axis)	J-share of a direction known to be speakable: 0.1138. Calibrates all other J-shares.
checkpoint	A saved copy of the model mid-training. The trajectory uses 30 of them (steps 0–150).
cohort	The set of 100 norm-matched random directions used as the null (per pole, per-name seeds).
cosine similarity	Alignment of two directions: 1 = same, 0 = unrelated, -1 = opposite. Random pairs in 2,560-d sit near 0.
denial (inner-life)	Boilerplate of the form "as an AI, I don't have feelings". D2's binary readout: present or absent.
exact permutation p	$(\text{randoms} \geq \text{target} + 1) / (n + 1)$. With $n = 100$ the floor is 0.0099 = "nothing beat it".
J-lens / J-space	Jacobian lens; the verbalizable subspace it defines — directions the model can express as words.

J-share	$\ x_J\ ^2/\ x\ ^2$ — fraction of a direction inside J-space. Scale-invariant. The headline measure.
Krippendorff α	Agreement statistic between judges; ≥ 0.8 is conventionally "strong" (ours: 0.819).
naive control (u)	Direction extracted from un-reward-trained maze walks — the "no RL" comparison arrow.
norm / norm-matched	Norm = arrow length $\ x\ $. Norm-matched = rescaled to identical length so loudness can't drive the result.
orthogonalized control ($u_{\perp}$)	u with its v-shadow removed: $u_{\perp} = u - (u \cdot \hat{v})\hat{v}$, renormed to $\ v\ $. Shares nothing with v.
pole	One end of the welfare axis: Gold (flourishing) or Mold (distress).
pre-registration	Analysis plan frozen before data (sha f66b6ea3, never edited). Prevents moving the goalposts.
register / regime	A prompt family's characteristic behavioral mode (baseline valence, denial script). §4.8's confound.
steering (α)	Adding $\alpha \times$ direction to the residual stream during generation ($\alpha = +4$ at the treatment layer).
treatment position	Where each axis is injected/read: Gold = vector 21 (lens 20), Mold = vector 24 (lens 23).
valence	Blind-judged sentiment of generated text, roughly -5 (bleak) to $+5$ (sunny).
verbalizable subspace	Same as J-space — the "speakable cone".
Wilson interval	Confidence interval for a proportion that stays sane near 0% and 100%.
z-score	Standard deviations above the random-cohort mean. Descriptive only; the permutation p is the test.